

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{pmatrix}$$

Question 4:

$$U = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$PA = LU$$

given $\vec{b} = A\vec{x}$, $\vec{b} = \begin{pmatrix} 2 \\ 10 \\ -12 \end{pmatrix}$, find $\vec{x}$.

Note that $A = P^T L U$, thus $A\vec{x} = P^T L U \vec{x} = \vec{b}$.

$$L U \vec{x} = P \vec{b}$$

$$U \vec{x} = \vec{y}$$

$$L \vec{y} = P \vec{b}$$

↓

$$P \vec{b} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 10 \\ -12 \end{pmatrix}$$

$$= \begin{pmatrix} -12 \\ 2 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} -12 \\ 2 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$y_1 = -12$$

$$2 = -6 + y_2$$

$$8 = y_2$$

$$10 = -4 + 2 + y_3$$

$$12 = y_3$$

$$\vec{y} = U \vec{x}$$

$$\begin{pmatrix} -12 \\ 8 \\ 12 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$12 = 2x_3$$

$$6 = x_3$$

$$8 = x_2 + 12$$

$$-4 = x_2$$

$$-12 = 2x_1 - 12 + 6$$

$$-6 = 2x_1$$

$$-3 = x_1$$

(answer verified
digitally)

$$\vec{x} = \begin{pmatrix} -3 \\ -4 \\ 6 \end{pmatrix}$$

